Supplementary Material: Communication as the Binding Constraint

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I. S1. Relation to the superseded version

An earlier manuscript ("Nested Federated Learning" / "TierFed" v2) reported results produced by code with the following defects, found in an internal audit and corrected here: (1) differential privacy was applied inside two strategies only and clipped the aggregate rather than each client's update, so FedAvg/FedPer/FedProx ran without noise and the reported budgets were invalid; the TierFed DP runs crashed on local-only rounds and the table was unsupported; (2) the SCAFFOLD baseline was FedAvg with doubled bit accounting; (3) the scalability table contained no TierFed runs and its values were not those of the logs; (4) the architecture sweep compared different depths across methods; (5) the imaging result used an undisclosed configuration on a binarised 97%-positive task; (6) the inversion attack targeted the fixed schedule at initialisation and was labelled TierFed; (7) the model described (93 inputs) was not the model trained (162 inputs); (8) repeated runs were not deterministic; (9) no patient-level deduplication was performed and one "missing specialty" client held 49% of the data; (10) the greedy controller reached only powers of two; (11) the fixed multi-rate schedule sent a fallback tier whenever no tier was due, which made its medium-tier period ineffective; (12) compressors were applied to raw weights instead of updates. Items (1)–(10) were found in an audit of the earlier manuscript and items (11)–(12) during the v3 runs; all affected runs were deleted and rerun. With these corrections the earlier central finding, that fixed multi-rate schedules lose about 0.05 AUROC to personalised baselines, does not reproduce and is retracted. Every item maps to a code change in the released repository (`ROADMAP.md`, section 0) and to a test.

II. S2. Derivation of Proposition 1

We follow the cooperative-SGD framework of Wang and Joshi [JMLR 2021, Thm. 1], which bounds the averaged gradient norm of local-update SGD in terms of the per-round deviation of client iterates from the virtual average. Write the shared parameter as $\theta = (\theta_g)_g$ with disjoint blocks. For site k and block g let $\theta_{k,g}^{(t)}$ be the iterate after t local steps since block g was last synchronised and $\bar{\theta}_g^{(t)}$ the (virtual) average across sites.

Lemma 1 (Blockwise deviation)**.** *Under A1–A3, for* $\eta \le c_0/(LK_{\max}\tau_{\max})$ *and any* $t \le K_g\tau_{\max}$,

$$\frac{1}{N}\sum_k \mathbb{E}\big\|\theta_{k,g}^{(t)} - \bar{\theta}_g^{(t)}\big\|^2 \;\le\; c\,\eta^2 t^2\,(\Gamma_g + \sigma_g^2) \;\le\; c\,\eta^2 K_g^2 \tau_{\max}^2 (\Gamma_g + \sigma_g^2).$$

Proof. Identical to Lemma 2 of Wang and Joshi restricted to the coordinates of block g: the deviation after t steps is a sum of t stochastic-gradient differences, each with second moment bounded by $\Gamma_g + \sigma_g^2$ by (A2)–(A3), and the smoothness term is absorbed by the step-size condition. The restriction to a block is valid because the gradient of block g is bounded blockwise by the same assumptions. □

The descent-lemma step of the cited theorem bounds $\mathbb{E}\|\nabla F(\bar{\theta}_r)\|^2$ by the progress term, the variance term and L^2 times the *total* deviation $\frac{1}{N}\sum_k \|\theta_k - \bar{\theta}\|^2$. Because the blocks are disjoint, the total deviation is the sum over g of the blockwise deviations, each bounded by the lemma with its own K_g; summing and averaging over rounds gives Proposition 1 of the main text with constants inherited from the cited theorem. We make no claim about the constants beyond their existence. Setting $K_g = 1$ for a single block returns the cited bound with τ local steps.

What the controller sees. By the same lemma the dispersion of the received slices of tier g after K_g rounds is of order $\eta^2 K_g^2 \tau^2 (\Gamma_g + \sigma_g^2)$; normalising by $\sum_j \|\bar{\theta}_j\|^2$ gives the scale-free $\hat{\Gamma}_g$ of Eq. (7) of the main text (the dispersion estimator). The controller therefore minimises $\sum_g K_g^2 \hat{\Gamma}_g$ where $\hat{\Gamma}_g$ already carries the factor K_g^2 of the interval that produced it; it is a feedback loop on a proxy, and its stability is observed (Fig. 4 of the main text), not proved.

III. S3. Differential-privacy accounting

Per-client clipping to C makes the ℓ_2 sensitivity of the sum $\sum_k \tilde{\Delta}_k$ to one client exactly C. The Gaussian mechanism with standard deviation σC satisfies $(\alpha, \alpha/(2\sigma^2))$-RDP for every $\alpha > 1$ [Mironov 2017, Prop. 7]. Composition over the R_{tx} rounds in which something is transmitted gives $(\alpha, R_{\mathrm{tx}}\alpha/(2\sigma^2))$-RDP, and (ε, δ)-DP follows with $\varepsilon = \min_\alpha\{R_{\mathrm{tx}}\alpha/(2\sigma^2) + \log(1/\delta)/(\alpha - 1)\}$ [Mironov 2017, Prop. 3]. Rounds with no transmission release nothing and are not charged. Full participation is used, so no subsampling amplification applies.

IV. S4. Per-seed tables

Every run's final metrics are in `results/tables/collected_all_raw.csv` in the released repository. Total runs at build time: 1045.

V. S5. SGD arm

The SGD arm (learning rate 5×10^{-2}) is reported in Table VI of the main text (baseline sweeps).

VI. S6. Excluded sites on the imaging benchmark

Number of sites whose patient-grouped held-out split contained a single class (identical for every method, per seed): **[RUN REQUIRED]** (read from `n_clients_single_class_final` in the logs when the imaging stage has run).